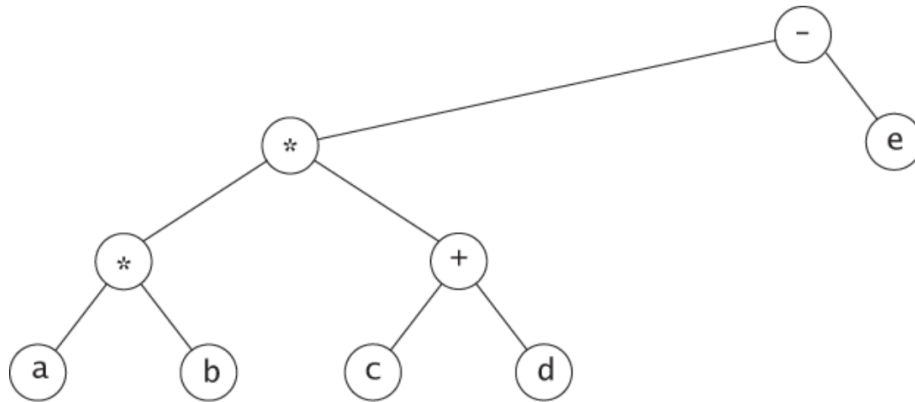


1. Give the prefix, infix, and postfix expressions that represent the tree below: **(15 points)**
(Please pay attention to the precedence of math operations and add parentheses when needed)



- a) Prefix:
- b) Infix:
- c) Postfix:

2. For each of the following cases, draw the result binary search tree: **(10 points)**
- a. After inserting 3, 1, 4, 6, 9, 2, 5, 7 into an initially empty binary search tree.

 - b. After deleting the root from the binary search tree of part a.

3. Draw all the rotations that you must perform and the final AVL tree after the **(10 points)** following elements are inserted in the given order starting from an empty tree: 1, 10, 5, 7, 3, 13, 6, 4, 8, 9

4. Given the following activities/tasks and their durations, draw an **activity-node graph** (not the event-node graph) from the **wake up** to the **leave** activities/tasks. (10 points)

Task Number	Task Name	Duration	Depends On
1	Choose clothes	3	9
2	Get Dressed	6.5	1, 8
3	Eat breakfast	6	5, 6, 7
4	Leave	1	2, 3
5	Make coffee	4.5	9
6	Make toast	2.0	9
7	Pour juice	0.5	9
8	Shower	8.5	9
9	Wake up	0	-

5. There are 6 cities: U, V, W, X, Y and Z. The cost of all possible routes between each pair of cities is shown in the table below. Note that " ∞ " means that there is no direct path between the two cities (cost is ∞). Your task is to determine a set of routes that will minimize the cost of connecting city Z to all other cities (not necessarily direct connections). **(15 points)**

	U	V	W	X	Y	Z
U	0	6	3	8	∞	∞
V	6	0	3	8	∞	∞
W	3	3	0	2	7	4
X	8	8	2	0	4	∞
Y	∞	∞	7	4	0	3
Z	∞	∞	4	∞	3	0

- Map the problem into a graph model (i.e. create a graph): what does a city represent?
What does a route represent?
- Describe an algorithm to achieve your task.
- Show how the algorithm works for this specific set of computers and costs. Please show your detailed work.